- The $MA(q)$ part of the process is stationary, and hence y_t is stationary if the AR part of the process is stationary, namely, if all the roots of the characteristic equation

$$\lambda^p - \phi_1 \lambda^{p-1} - \ldots - \phi_p = 0$$

lie inside the unit circle.

- $\phi(L)$ and $\theta(L)$ have no common roots; otherwise, the $ARMA$ model is unidentified and the order (p, q) can be reduced. By factorization,

$$y_t = \frac{\theta(L)}{\phi(L)}\varepsilon_t = \frac{(1-\eta_1 L)(1-\eta_2 L)\cdots(1-\eta_q L)}{(1-\lambda_1 L)(1-\lambda_2 L)\cdots(1-\lambda_p L)}\varepsilon_t.$$

If there is a pair $\eta_i = \lambda_j$, then the two factors $(1-\eta_i L)$ and $(1-\lambda_j L)$ cancel out and the model reduces to $ARMA(p-1, q-1)$.

- The ACF of an $ARMA(p, q)$ process displays the shape of that of an $AR(p)$ process for $h \geq \max(p, q+1)$.

- Three representations for an $ARMA$ model:

  - (i) $ARMA(p, q)$ representation: parsimonious parameterization; useful in parameter estimation and multistep-ahead forecasting

  - (ii) MA representation: If the $ARMA$ process given by (16) is stationary, then there exists constants $\{\psi_i\}$ such that $\sum_{i=0}^{\infty} |\psi_i| < \infty$ and $y_t = \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}$, for all t.

    * The MA representation explicitly shows the impact of the past shock ε_{t-i} $(i > 0)$ on the current outcome y_t. The coefficients $\{\psi_i\}$ are known as the *impulse response function* of the $ARMA$ model.

    * ψ_i decays exponentially as i increases, and accordingly the shock ε_{t-i} will not have a permanent impact on y_t.

  - (iii) AR representation: easy to estimate; useful in forecasting. Can we rewrite an

$ARMA$ process as a converging infinite order AR model?

$$y_t = \pi_1 y_{t-1} + \pi_2 y_{t-2} + \pi_3 y_{t-3} + \ldots + \varepsilon_t$$

Definition 6.1. (invertibility) The $ARMA$ process given by (16) is said to be *invertible* if there exits constants $\{\pi_i\}$ such that $\sum_{i=0}^{\infty} |\pi_i| < \infty$ and $\varepsilon_t = \sum_{i=0}^{\infty} \pi_i y_{t-i}$ for all t. Invertibility is equivalent to the condition that all the roots of

$$\theta(z) = 1 + \theta_1 z + \ldots + \theta_q z^q = 0$$

fall *outside* the unit circle.

- An invertible $ARMA$ process is a linear combination of the current shock and a weighted average of the past values, where the weights decay exponentially for more remote past values.
- Intuitively, π_i decays to zero as i increases because the impact of remote lags, y_{t-i}, on y_t and the current shock should diminish as i increases.
- The notion of invertibility can be motivated by the following example. Consider the $MA(1)$ process: $y_t = \varepsilon_t + \theta \varepsilon_{t-1}$. We have shown that

$$\rho_0 = 1, \quad \rho_1 = \frac{\theta}{1+\theta^2}, \quad \rho_h = 0, \text{ for } h > 1.$$

Note that ρ_h remains unchanged if θ is replaced with θ^{-1}. Moreover,

$$\gamma_0 = \left(1+\theta^2\right)\sigma^2, \quad \gamma_1 = \theta\sigma^2, \quad \gamma_h = 0, \text{ for } h > 1.$$

Clearly, replacing (θ, σ^2) with $(\theta^{-1}, \theta^2\sigma^2)$ leads to the same γ_h. For instance, the two

models

$$y_t = \varepsilon_t + \frac{1}{4}\varepsilon_{t-1},\ \varepsilon_t \sim WN\left(0, 16\right),$$
$$y_t = u_t + 4u_{t-1},\ u_t \sim WN\left(0, 1\right),$$

are indistinguishable since we only observe y_t but not the noise. Therefore, MA models exhibit non-uniqueness in their representation regarding the autocorrelation and autocovariance functions.

- We prefer the invertible representations in practice because ε_t can be expressed in terms of the past values of y. If a process is noninvertible, then ε_t is expressed in terms of all future values of y.

- *Exercise*: Determine whether the following processes are stationary and invertible.
  - (i) $y_t = 0.7y_{t-1} + \varepsilon_t + 1.25\varepsilon_{t-1}$
  - (ii) $y_t = 0.4y_{t-1}+0.45y_{t-2}+\varepsilon_t+\varepsilon_{t-1}+0.25\varepsilon_{t-2}$ (Is this process really an $ARMA\left(2,2\right)$?)

Definition 6.2. (linear process) A time series $\{y_t\}$ is said to be a *linear process* if it can be written as

$$y_t = \mu + \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i},$$

for all t, where $\{\varepsilon_t\} \sim WN\left(0, \sigma^2\right)$ and $\{\psi_i\}$ is a sequence of constants such that $\sum_{i=0}^{\infty} \psi_i^2 < \infty$.

- Some books define a linear process with the condition $\sum_{i=0}^{\infty} |\psi_i| < \infty$, or without the stationarity condition.

Theorem 6.1. (Wold's decomposition, aka Wold representation theorem) Any trend-stationary process $\{y_t\}$ can be represented in the form of

$$y_t = d_t + \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}, \tag{17}$$

where

1. $\{d_t\}$ is a deterministic component.
2. $\{\varepsilon_t\}$ is a serially uncorrelated process defined by innovations in y_t:

$$\varepsilon_t \equiv y_t - P\left(y_t | y_{t-1}, y_{t-2}, \ldots\right),$$

   where $P\left(y_t | y_{t-1}, y_{t-2}, \ldots\right)$ denotes projection, i.e., the fitted value of a linear regression of y_t on $y_{t-1}, y_{t-2}, \ldots$, [5] $\varepsilon_t \sim WN\left(0, \sigma^2\right)$, and $E\left(\varepsilon_t d_s\right) = 0$ for all t, s.
3. $\psi_0 = 1$ and $\sum_{i=0}^{\infty} \psi_i^2 < \infty$. The MA polynomial is invertible.

- Proofs are given in Sargent (1987, pp. 286–290), and Brockwell and Davis (1991, pp. 188–189). A sketch can be found in this note.
- Remarks:
  - This theorem states that *any* covariance stationary time series has a linear ARMA representation!
  - $\{\varepsilon_t\}$ are the linear forecast errors of $\{y_t\}$. By construction, $E\left(\varepsilon_t y_{t-i}\right) = 0$ for $i > 0$. $\{\varepsilon_t\}$ are referred to as the *Wold innovations* or *Wold shocks*, which need not be the "true" shocks to the system.
  - $\{\psi_i\}$ and $\{\varepsilon_t\}$ are unique—the Wold representation is the unique *linear* representation where the shocks are *linear* forecast errors.

[5] Linear regression can be viewed as a projection. See this article for a discussion.

– $\{\varepsilon_t\}$ need not be i.i.d. or normally distributed.

7 Estimation of Stationary Processes

7.1 Estimation of Mean and Autocovariances

- Suppose that $\{y_t\}_{t=1}^T$ is a covariance stationary process with population mean $\mu = E(y_t)$. The sample mean is given by

$$\hat{\mu} = \bar{y}_T = \frac{y_1 + y_2 + \ldots + y_T}{T}.$$

It is easily seen that $\bar{y}_T$ is an unbiased estimator of μ. Is it a consistent estimator?

- A covariance stationary process is said to be *ergodic for the mean* (or *ergodic in mean*) if $\bar{y}_T$ converges in probability to μ, as $T \to \infty$.

- A sufficient condition for $\bar{y}_T$ to be consistent of μ is $\lim_{T\to\infty} Var(\bar{y}_T) = 0$. Note that

$$\begin{aligned} Var(\bar{y}_T) &= \frac{1}{T^2}\left[T\gamma_0 + 2(T-1)\gamma_1 + 2(T-2)\gamma_2 + \ldots + 2\gamma_{T-1}\right] \\ &= \frac{1}{T}\left[\gamma_0 + 2\sum_{h=1}^{T-1}\left(1 - \frac{h}{T}\right)\gamma_h\right]. \end{aligned}$$

To ensure $Var(\bar{y}_T) \to 0$, as $T \to \infty$, it is sufficient that (noting that $\gamma_0 < \infty$)

$$\lim_{T\to\infty}\left[\sum_{h=1}^{T-1}\left(1 - \frac{h}{T}\right)\gamma_h\right] < \infty,$$

which is satisfied if the autocovariances are absolutely summable, i.e., $\sum_{h=0}^{\infty}|\gamma_h| < \infty$.[6] That is, if the autocovariances for a weakly stationary process satisfy the absolute summability condition, then $\{y_t\}$ is ergodic for the mean.

- Suppose that $\{y_t\}_{t=1}^T$ is a covariance stationary process with finite fourth-order moment. If $\{y_t\}$ has absolutely summable autocovariances, then $\hat{\gamma}_T(h) \to_p \gamma_h$, for a fixed h as $T \to \infty$,

[6] A less stringent condition, $\lim_{T\to\infty}\sum_{h=1}^{T-1}\gamma_h < \infty$, is sufficient for a consistent estimation of the mean.

where

$$\hat{\gamma}_T(h) = \frac{\sum_{t=h+1}^{T}(y_t - \bar{y}_T)(y_{t-h} - \bar{y}_T)}{T},$$

and $\{y_t\}$ is said to be *ergodic in variance* or *ergodic in second moments.*

7.2 Estimation of AR Models

Consider the $AR(p)$ process:

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t, \quad \varepsilon_t \sim IID\left(0, \sigma^2\right).$$

Suppose that the observations $(y_1, y_2, \ldots, y_T)$ are available and the order p is known. Let $\boldsymbol{\phi} = (\phi_1, \phi_2, \ldots, \phi_p)'$ and $\boldsymbol{\theta} = (\boldsymbol{\phi}', \sigma^2)'$.

7.2.1 Yule-Walker Estimation

Recall that the Yule-Walker (YW) equations are

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{p-2} \\ \vdots & \vdots & \cdots & \vdots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots & \gamma_0 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{pmatrix},$$

which can be written more compactly as

$$\boldsymbol{\Gamma}_p \boldsymbol{\phi} = \boldsymbol{\gamma}_p.$$

Let $\hat{\boldsymbol{\Gamma}}_p$ and $\hat{\boldsymbol{\gamma}}_p$ denote the estimators of $\boldsymbol{\Gamma}_p$ and $\boldsymbol{\gamma}_p$, obtained by replacing the autocovariances with the sample autocovariances, $\hat{\gamma}_h = \frac{1}{T}\sum_{t=h+1}^{T} y_t y_{t-h}$. Then the YW estimators of $\boldsymbol{\phi}$ are given by

$$\hat{\boldsymbol{\phi}}_{YW} = \hat{\boldsymbol{\Gamma}}_p^{-1} \hat{\boldsymbol{\gamma}}_p.$$

Also recall $\gamma_0 = \phi_1\gamma_1 + \phi_2\gamma_2 + \ldots + \phi_p\gamma_p + \sigma^2$, and hence

$$\hat{\sigma}^2 = \hat{\gamma}_0 - \hat{\boldsymbol{\phi}}'_{YW}\hat{\boldsymbol{\gamma}}_p.$$

- *Method of moments*: The parameters are solved by equating the sample moments and the population moments. The number of moments equals to the number of parameters to be estimated.
- $\hat{\boldsymbol{\phi}}$ is asymptotically normal, $\sqrt{T}\left(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}\right) \sim N\left(\mathbf{0}, \sigma^2\boldsymbol{\Gamma}_p^{-1}\right)$, as $T \to \infty$, and $\hat{\sigma}^2 \to_p \sigma^2$.

7.2.2 Maximum Likelihood and Least Squares Estimations

AR(1) process

- First consider the $AR(1)$ process: $y_t = \phi y_{t-1} + \varepsilon_t$, where $|\phi| < 1$, $\varepsilon_t \sim IIDN\left(0, \sigma^2\right)$. Also note that the joint densities can be written as

$$\begin{aligned} f(y_1, y_2) &= f(y_2|y_1) f(y_1) \\ f(y_1, y_2, y_3) &= f(y_3|y_2, y_1) f(y_2, y_1) \\ &\ldots \\ f(y_1, y_2, \ldots, y_T) &= f(y_1) f(y_2|y_1) f(y_3|y_2, y_1) \ldots f(y_T|y_{T-1}, y_{T-2}, \ldots, y_1) \end{aligned}$$

  But $f(y_t|y_{t-1}, y_{t-2}, \ldots, y_1) = f(y_t|y_{t-1})$, for $t = 2, 3, \ldots, T$, and hence

$$f(y_1, y_2, \ldots, y_T) = f(y_1) f(y_2|y_1) f(y_3|y_2) \ldots f(y_T|y_{T-1}).$$

  The *likelihood function* is the joint density of $y_1, y_2, \ldots, y_T$ viewed as a function of the parameters, $\boldsymbol{\theta} = \left(\phi, \sigma^2\right)'$, and can be written as

$$L(\boldsymbol{\theta}) = f(y_1; \boldsymbol{\theta}) \times \prod_{t=2}^{T} f(y_t|y_{t-1}; \boldsymbol{\theta}).$$

Suppose the process has started a long time ago, we have $y_1 \sim N\left(0, \frac{\sigma^2}{1-\phi^2}\right)$ and $y_t|y_{t-1} \sim N\left(\phi y_{t-1}, \sigma^2\right)$, that is,

$$f\left(y_1;\boldsymbol{\theta}\right) = \frac{1}{\sqrt{2\pi\sigma^2/\left(1-\phi^2\right)}} \exp\left[-\frac{\left(1-\phi^2\right)y_1^2}{2\sigma^2}\right],$$

$$f\left(y_t|y_{t-1};\boldsymbol{\theta}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(y_t-\phi y_{t-1}\right)^2}{2\sigma^2}\right], \text{ for } t = 2, 3, \ldots, T.$$

Therefore,

$$\begin{aligned}\log L\left(\boldsymbol{\theta}\right) &= \log f\left(y_1;\boldsymbol{\theta}\right) + \sum_{t=2}^{T} \log f\left(y_t|y_{t-1};\boldsymbol{\theta}\right) \\ &= \left[-\frac{1}{2}\log\left(\frac{2\pi\sigma^2}{1-\phi^2}\right) - \left(\frac{1-\phi^2}{2\sigma^2}\right)y_1^2\right] + \left[-\frac{(T-1)}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=2}^{T}\left(y_t-\phi y_{t-1}\right)^2\right].\end{aligned} \tag{18}$$

- The *maximum likelihood estimator* (MLE) of $\boldsymbol{\theta}$ is $\hat{\boldsymbol{\theta}}_{MLE} = \text{argmax}_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} L\left(\boldsymbol{\theta}\right)$, or equivalently, $\hat{\boldsymbol{\theta}}_{MLE} = \text{argmax}_{\boldsymbol{\theta}\in\boldsymbol{\Theta}} \log L\left(\boldsymbol{\theta}\right)$.

- If $|\phi| < 1$, the effect of the density of the initial observation will diminish as $T \to \infty$. One may ignore $\log f\left(y_1;\boldsymbol{\theta}\right)$ only if $|\phi| < 1$ and T is sufficiently large. If $\log f\left(y_1;\boldsymbol{\theta}\right)$ is ignored,

$$\log L\left(\boldsymbol{\theta}\right) \approx -\frac{(T-1)}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=2}^{T}\left(y_t-\phi y_{t-1}\right)^2 \tag{19}$$

  the estimator that maximizes the approximate log-likelihood function, (19), is called the *conditional maximum likelihood estimator* (CMLE). By contrast, the full log-likelihood given by (18) is referred to as the exact log-likelihood, and its maximizer is known as the exact MLE.

- The CMLE of $\boldsymbol{\phi}$ can be computed by

$$\hat{\phi}_{LS} = \text{argmin}_{\phi}\left[\sum_{t=2}^{T}\left(y_t-\phi y_{t-1}\right)^2\right],$$

or

$$\hat{\phi}_{LS} = \frac{\sum_{t=2}^{T} y_t y_{t-1}}{\sum_{t=2}^{T} y_{t-1}^2},$$

which is the OLS coefficient of the regression of y_t on y_{t-1}, for $t = 2, 3, \ldots, T$. Note that the least squares regression must also include an intercept if y_t has a non-zero mean.

AR(p) process

- Now consider the stationary $AR(p)$ process. The likelihood function can be written as

$$L(\boldsymbol{\theta}) = f(y_1, y_2, \ldots, y_p) \times \prod_{t=p+1}^{T} f(y_t | y_{t-1}, y_{t-2}, \ldots, y_{t-p}),$$

and the log-likelihood function is given by

$$\log L(\boldsymbol{\theta}) = \log f(y_1, y_2, \ldots, y_p) + \sum_{t=p+1}^{T} \log f(y_t | y_{t-1}, y_{t-2}, \ldots, y_{t-p}).$$

Under $\varepsilon_t \sim IIDN(0, \sigma^2)$, we have

$$f(y_t | y_{t-1}, y_{t-2}, \ldots, y_{t-p}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(y_t - \sum_{i=1}^{p} \phi_i y_{t-i}\right)^2}{2\sigma^2} \right], \text{ for } t > p,$$

and then

$$\log L(\boldsymbol{\theta}) = \log f(y_1, y_2, \ldots, y_p) - \frac{(T-p)}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2} \sum_{t=p+1}^{T} \left(y_t - \sum_{i=1}^{p} \phi_i y_{t-i} \right)^2.$$

When the $AR(p)$ process is stationary, the first term on the RHS is asymptotically negligible (for sufficiently large T), and the log-likelihood function can be approximated by

$$\log L(\boldsymbol{\theta}) \approx -\frac{(T-p)}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2} \sum_{t=p+1}^{T} \left(y_t - \sum_{i=1}^{p} \phi_i y_{t-i} \right)^2.$$

- The CMLE of $\boldsymbol{\phi}$ can be computed by

$$\hat{\boldsymbol{\phi}}_{LS} = \text{argmin}_{\boldsymbol{\phi}} \left[\sum_{t=p+1}^{T} \left(y_t - \sum_{i=1}^{p} \phi_i y_{t-i} \right)^2 \right],$$

or

$$\hat{\boldsymbol{\phi}}_{LS} = \left(\sum_{t=p+1}^{T} \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \sum_{t=p+1}^{T} \mathbf{x}_t y_t = \left(\mathbf{X}_p' \mathbf{X}_p \right)^{-1} \mathbf{X}_p' \mathbf{y}_p,$$

where $\mathbf{x}_t = (y_{t-1}, y_{t-2}, \ldots, y_{t-p})'$ for $t > p$,

$$\mathbf{X}_p = \begin{pmatrix} y_p & y_{p-1} & \cdots & y_1 \\ y_{p+1} & y_p & \cdots & y_2 \\ \vdots & \vdots & & \vdots \\ y_{T-1} & y_{T-2} & \cdots & y_{T-p} \end{pmatrix}, \quad \mathbf{y}_p = \begin{pmatrix} y_{p+1} \\ y_{p+2} \\ \vdots \\ y_T \end{pmatrix}.$$

That is, $\hat{\boldsymbol{\phi}}_{LS}$ can be computed by the least squares regression of y_t on $y_{t-1}, y_{t-2}, \ldots, y_{t-p}$, for $t = p+1, p+2, \ldots, T$. In addition, the CMLE for σ^2 is given by

$$\hat{\sigma}^2 = \frac{\left(\mathbf{y}_p - \mathbf{X}_p \hat{\boldsymbol{\phi}}_{LS} \right)' \left(\mathbf{y}_p - \mathbf{X}_p \hat{\boldsymbol{\phi}}_{LS} \right)}{T - p}.$$

- If y_t has a non-zero mean, i.e., $y_t = \phi_0 + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t$ and $\boldsymbol{\phi} = (\phi_0, \phi_1, \ldots, \phi_p)'$, then the least squares regression must also include an intercept, namely, in the above expressions $\mathbf{x}_t = (1, y_{t-1}, y_{t-2}, \ldots, y_{t-p})'$, and $\mathbf{X}_p$ includes a column of ones.

- Although the CMLE was derived assuming $\{y_t\}$ is Gaussian, it consistently estimates the *AR* parameters without Gaussianity as long as the conditional mean is correctly specified.

- For pure AR processes, the YW, ML, and LS estimators give same results as $T \to \infty$. For large T, the YW/LS estimators are preferred. For small T and especially if the series is Gaussian, ML estimation has better performance.

7.3 Estimation of MA Models

- The exact likelihood function for an MA process is complicated since $\{\varepsilon_t\}$ are unobserved. However, the calculation of the likelihood function can be considerably simplified if we condition on initial values for the ε's. The idea is similar to conditioning on the initial values for the y's when we compute the conditional likelihood function for an AR model.

- We illustrate the idea for the $MA(1)$ process: $y_t = \varepsilon_t + \theta\varepsilon_{t-1}$, where $\varepsilon_t \sim IIDN\left(0, \sigma^2\right)$. Let $\boldsymbol{\theta}=\left(\theta, \sigma^2\right)'$ denote the parameters to be estimated. Suppose that we know for certain that $\varepsilon_0 = 0$. Then

$$\begin{aligned} \varepsilon_1 &= y_1, \\ \varepsilon_2 &= y_2 - \theta\varepsilon_1, \\ &\vdots \\ \varepsilon_t &= y_t - \theta\varepsilon_{t-1}, \\ &\vdots \end{aligned} \tag{20}$$

That is, the full sequence of $\{\varepsilon_t\}_{t=1}^T$ can be calculated by iterating on (20), for $t = 1, 2, \ldots, T$, starting from $\varepsilon_0 = 0$. It follows that

$$f\left(y_1|\varepsilon_0 = 0; \boldsymbol{\theta}\right) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\varepsilon_1^2}{2\sigma^2}\right),$$

$$f\left(y_2|y_1, \varepsilon_0 = 0; \boldsymbol{\theta}\right) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{\left(y_2 - \theta\varepsilon_1\right)^2}{2\sigma^2}\right] = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\varepsilon_2^2}{2\sigma^2}\right),$$

$$f\left(y_3|y_2, y_1, \varepsilon_0 = 0; \boldsymbol{\theta}\right) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{\left(y_3 - \theta\varepsilon_2\right)^2}{2\sigma^2}\right] = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\varepsilon_3^2}{2\sigma^2}\right),$$

$$\vdots$$

$$f\left(y_t|y_{t-1}, y_{t-2}, \ldots, y_1, \varepsilon_0 = 0; \boldsymbol{\theta}\right) = f\left(y_t|\varepsilon_{t-1}; \boldsymbol{\theta}\right) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{\varepsilon_t^2}{2\sigma^2}\right).$$

Then the conditional likelihood function is given by

$$L\left(\boldsymbol{\theta}\right) = f\left(y_1, y_2, \ldots, y_T|\varepsilon_0 = 0; \boldsymbol{\theta}\right) = f\left(y_1|\varepsilon_0 = 0; \boldsymbol{\theta}\right)\prod_{t=2}^{T} f\left(y_t|y_{t-1}, y_{t-2}, \ldots, y_1, \varepsilon_0 = 0; \boldsymbol{\theta}\right),$$

and the conditional log-likelihood is

$$\log L(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\left(\sigma^2\right) - \sum_{t=1}^{T}\frac{\varepsilon_t^2}{2\sigma^2}.$$

The CMLE of $\boldsymbol{\theta}$ can be obtained by maximizing $\log L(\boldsymbol{\theta})$ through numerical optimization—for a given value of $\boldsymbol{\theta}$, we can calculate $\{\varepsilon_t\}$ by iterating on (20), and then $\log L(\boldsymbol{\theta})$ is a function of the sum of squares of these ε's.

- Iterating on (20) from an arbitrary starting value of ε_0 results in

$$\varepsilon_t = y_t - \theta y_{t-1} + \theta^2 y_{t-2} - \ldots + (-1)^{t-1}\theta^{t-1} y_1 + (-1)^t \theta^t \varepsilon_0.$$

If $|\theta|$ is much smaller than one and the sample size is reasonably large, the effect of imposing $\varepsilon_0 = 0$ will quickly diminish, and the conditional likelihood is a good approximation to the unconditional likelihood. However, if $|\theta| > 1$, the effect of imposing $\varepsilon_0 = 0$ will accumulate over time.

- For the $MA(q)$ process, $y_t = \sum_{i=0}^{q}\theta_i\varepsilon_{t-i}$ with $\theta_0 = 1$, the conditional log-likelihood function assuming that the first q values for ε are all zero ($\varepsilon_0 = \varepsilon_{-1} = \ldots = \varepsilon_{-q+1} = 0$) is given by

$$\log f(y_1, y_2, \ldots, y_T | \varepsilon_0 = \varepsilon_{-1} = \ldots = \varepsilon_{-q+1} = 0; \boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\left(\sigma^2\right) - \sum_{t=1}^{T}\frac{\varepsilon_t^2}{2\sigma^2},$$

where $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_q, \sigma^2)'$. Note that the above approximation to the full likelihood should be used only if the process is invertible.

7.4 Estimation of ARMA Models

- Consider the zero-mean stationary $ARMA(p,q)$ process, $\{y_t\}$, where the parameters are denoted by $\boldsymbol{\phi} = (\phi_1, \phi_2, \ldots, \phi_p)'$, $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_q)'$, and σ^2. The estimation of $ARMA$ models is considerably more complicated since

– $\{\varepsilon_t\}$ is unobserved, and hence we cannot use the conditional distribution

$$y_t|y_{t-1},\ldots,y_{t-p},\varepsilon_{t-1},\ldots,\varepsilon_{t-q} \sim N\left(\phi_1 y_{t-1}+\ldots+\phi_p y_{t-p}+\varepsilon_t+\theta_1\varepsilon_{t-1}+\ldots+\theta_q\varepsilon_{t-q},\sigma^2\right)$$

– ε_t depends on the entire past $\{y_{t-1},y_{t-2},\ldots,y_1\}$ (under invertibility)

- We will only briefly explain the basic idea of the maximum likelihood estimation.[7] Let $\boldsymbol{\beta}=(\boldsymbol{\phi}',\boldsymbol{\theta}')'$, $\mathbf{y}=(y_1,y_2,\ldots,y_T)'$, and $\boldsymbol{\Gamma}_T\left(\boldsymbol{\beta},\sigma^2\right)$ be the $T\times T$ covariance matrix of $\mathbf{y}$, namely, $\boldsymbol{\Gamma}_T\left(\boldsymbol{\beta},\sigma^2\right)=E\left(\mathbf{y}\mathbf{y}'\right)$. The exact likelihood function is

$$L\left(\boldsymbol{\beta},\sigma^2\right)=\frac{1}{(2\pi)^{T/2}\left|\boldsymbol{\Gamma}_T\left(\boldsymbol{\beta},\sigma^2\right)\right|^{1/2}}\exp\left[-\frac{1}{2}\mathbf{y}'\boldsymbol{\Gamma}_T^{-1}\left(\boldsymbol{\beta},\sigma^2\right)\mathbf{y}\right],$$

and

$$\log L\left(\boldsymbol{\beta},\sigma^2\right)=-\frac{T}{2}\log\left(2\pi\right)-\frac{1}{2}\log\left|\boldsymbol{\Gamma}_T\left(\boldsymbol{\beta},\sigma^2\right)\right|-\frac{1}{2}\mathbf{y}'\boldsymbol{\Gamma}_T^{-1}\left(\boldsymbol{\beta},\sigma^2\right)\mathbf{y}.$$

The key is to express the autocovariances as functions of unknown parameters. The maximizer of the above log-likelihood function is the (exact) MLE of the $ARMA$ model.

- For example, for the $MA\left(1\right)$ model, $y_t=\varepsilon_t+\theta\varepsilon_{t-1}$, $|\theta|<1$, $\varepsilon_t\sim IIDN\left(0,\sigma^2\right)$,

$$\boldsymbol{\Gamma}_T\left(\theta,\sigma^2\right)=\sigma^2\begin{pmatrix}1+\theta^2 & \theta & 0 & \ldots & 0\\ \theta & 1+\theta^2 & \theta & \ldots & 0\\ \vdots & \vdots & \vdots & \vdots & \vdots\\ 0 & \ldots & \ldots & \theta & 1+\theta^2\end{pmatrix}.$$

- Recall that the approximation to the likelihood function for an AR process conditions on initial values of y's, whereas the approximation to the likelihood function for an MA process conditions on initial values of ε's. For an $ARMA\left(p,q\right)$ process, if we take initial observations $\mathbf{y}_0=(y_0,y_{-1},\ldots,y_{-p+1})'$ and $\boldsymbol{\varepsilon}_0=(\varepsilon_0,\varepsilon_{-1},\ldots,\varepsilon_{-q+1})'$ as given, then

[7] See Hamilton (1994), Ch. 5, for details.

$\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T\}$ can be computed from $\{y_1, y_2, \ldots, y_T\}$ by iterating on

$$\varepsilon_t = y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \ldots - \phi_p y_{t-p} - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \ldots - \theta_q \varepsilon_{t-q}, \qquad (21)$$

for $t = 1, 2, \ldots, T$, and then the conditional log-likelihood function is

$$\log f\left(y_T, y_{T-1}, \ldots, y_1 | \mathbf{y}_0, \boldsymbol{\varepsilon}_0; \boldsymbol{\beta}, \sigma^2\right) = -\frac{T}{2} \log\left(2\pi\sigma^2\right) - \sum_{t=1}^{T} \frac{\varepsilon_t^2}{2\sigma^2}.$$

One may set initial y's and ε's to their expected values, or alternatively let $\varepsilon_p = \varepsilon_{p-1} = \ldots = \varepsilon_{p-q+1} = 0$ and $y_1, y_2, \ldots, y_p$ be the observed values, and start the iteration (21) at $t = p + 1$. Using the latter option, the conditional likelihood function is

$$\begin{aligned} &\log f\left(y_T, y_{T-1}, \ldots, y_{p+1} | y_p, \ldots, y_1, \varepsilon_p = \varepsilon_{p-1} = \ldots = \varepsilon_{p-q+1} = 0; \boldsymbol{\beta}, \sigma^2\right) \\ &= -\frac{T-p}{2} \log\left(2\pi\sigma^2\right) - \sum_{t=p+1}^{T} \frac{\varepsilon_t^2}{2\sigma^2}. \end{aligned}$$

It is worth emphasizing that the approximation to the full likelihood should be used only if the process is invertible.

- Under certain regularity conditions, the MLE and CMLE are consistent and have the same asymptotic normal distribution: $\sqrt{T}\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right) \to_d N\left(\mathbf{0}, \sigma^2 \boldsymbol{\Gamma}_{p+q}^{-1}\right)$, as $T \to \infty$, where

$$\boldsymbol{\Gamma}_{p+q} = \begin{pmatrix} \boldsymbol{\Gamma}_{\phi\phi} & \boldsymbol{\Gamma}_{\phi\theta} \\ \boldsymbol{\Gamma}_{\theta\phi} & \boldsymbol{\Gamma}_{\theta\theta} \end{pmatrix},$$

with $\boldsymbol{\Gamma}_{\phi\phi} = [\gamma_u(i-j)]$, $i, j = 1, 2, \ldots, p$, for $AR(p)$ model $\phi(L) u_t = \varepsilon_t$,
$\boldsymbol{\Gamma}_{\theta\theta} = [\gamma_v(i-j)]$, $i, j = 1, 2, \ldots, q$, for $AR(q)$ model $\theta(L) v_t = \varepsilon_t$,
$\boldsymbol{\Gamma}_{\phi\theta} = \boldsymbol{\Gamma}'_{\theta\phi} = [\gamma_{uv}(i-j)]$, $i = 1, 2, \ldots, p$, $j = 1, 2, \ldots, q$, is cross-covariance $\gamma_{uv}(h)$.

- For example (assuming stationarity and invertibility):
$AR(1)$: $y_t = \phi y_{t-1} + \varepsilon_t$

$$\hat{\phi} \sim N\left[\phi, T^{-1}\left(1 - \phi^2\right)\right]$$

$AR(2)$: $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$

$$\begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} \sim N\left[\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}, \frac{1}{T} \begin{pmatrix} 1-\phi_2^2 & -\phi_1(1+\phi_2) \\ -\phi_1(1+\phi_2) & 1-\phi_2^2 \end{pmatrix} \right]$$

$MA(1)$: $y_t = \varepsilon_t + \theta \varepsilon_{t-1}$

$$\hat{\theta} \sim N\left[\theta, T^{-1}\left(1-\theta^2\right)\right]$$

$MA(2)$: $y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$

$$\begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{pmatrix} \sim N\left[\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \frac{1}{T} \begin{pmatrix} 1-\theta_2^2 & \theta_1(1-\theta_2) \\ \theta_1(1-\theta_2) & 1-\theta_2^2 \end{pmatrix} \right]$$

$ARMA(1,1)$: $y_t = \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$

$$\begin{pmatrix} \hat{\phi} \\ \hat{\theta} \end{pmatrix} \sim N\left[\begin{pmatrix} \phi \\ \theta \end{pmatrix}, \frac{1}{T} \frac{1+\phi\theta}{(\phi+\theta)^2} \begin{pmatrix} (1-\phi^2)(1+\phi\theta) & -(1-\theta^2)(1-\phi^2) \\ -(1-\theta^2)(1-\phi^2) & (1-\theta^2)(1+\phi\theta) \end{pmatrix} \right]$$

- In finite samples, the MLE and CMLE are generally not equal and may differ by a notable amount if $\{y_t\}$ is close to being non-invertible or non-stationary.

8 The Box-Jenkins Method

- Box-Jenkins (1976) proposed a three-step (iterative) method to apply ARMA models to time series data:
  - **Step 1. Model identification:** use the data and related information to help select a subclass of model: examine time plot of the series (check trend/cycle/stationarity, identify any unusual observations such as outliers, missing values and structural breaks), examine the sample ACF/PACF